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Self-pacing increases critical power and improves performance during severe-intensity exercise

Original Article

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ABSTRACT

The parameters of the power-duration relationship for severe-intensity exercise (i.e. the critical power, CP, and the curvature constant, W′) are related to the kinetics of pulmonary O_2 uptake ($\dot{V}O_2$), which may be altered by pacing strategy. We tested the hypothesis that the CP would be higher when derived from a series of self-paced time-trials (TT) than when derived from the conventional series of constant work-rate (CWR) exercise tests. Ten male subjects (21.5 ± 1.9 y, 75.2 ± 11.5 kg) completed 3-4 CWR and 3-4 TT prediction trial protocols on a cycle ergometer for the determination of the CP and W′. The CP derived from the TT protocol (265 ± 44 W) was greater (P<0.05) than the CP derived from the CWR protocol (250 ± 47 W), while the W′ was not different between protocols (TT: 18.1 ± 5.7 kJ, CWR: 20.6 ± 7.4 kJ, P>0.05). The $\dot{V}O_2$ mean response time (MRT) was shorter during the TTs than the CWR trials (TT: 34 ± 16, CWR: 39 ± 19 s, P<0.05). The CP was correlated with the total O_2 consumed in the first 60 s across both protocols (r=0.88, P<0.05, n=20). These results suggest that, in comparison with the conventional CWR exercise protocol, a self-selected pacing strategy enhances CP and therefore improves severe-intensity exercise performance. The greater CP during TT compared to CWR exercise has important implications for performance prediction, suggesting that TT completion times may be overestimated by CP and W′ parameters derived from CWR protocols.

Key words: Power-duration relationship; critical power; W′; $\dot{V}O_2$ mean response time; time trial; severe-intensity exercise

INTRODUCTION

The tolerable duration of exercise decreases in a hyperbolic fashion as power output (P) increases within the severe-intensity exercise domain (Jones et al. 2010; Morton 2006). This hyperbolic power-duration relationship is characterised by a power-asymptote, termed critical power (CP), and a curvature constant, termed W′. The CP denotes the highest work rate that can be sustained without a progressive loss of intramuscular and systemic homeostasis (Jones et al. 2008a; Poole et al. 1988) and is strongly associated with endurance exercise performance (Black et al. 2014, Smith et al. 1999). The W′ represents a fixed amount of work that can be performed above CP before exhaustion (Moritani et al. 1981; Poole et al. 1988). During exercise above CP, where pulmonary O_2 uptake ($\dot{V}O_2$) continues to increase until its maximum ($\dot{V}O_2max$) is achieved (Poole et al. 1988), exercise tolerance is predictable according to the power-duration relationship, based on the premise that the size of the W′ remains constant irrespective of its rate of expenditure (Chidnok et al. 2013, Morton 2006). Knowledge of the CP and W′ is therefore of considerable importance for understanding the limitations to high-intensity exercise tolerance and for predicting athletic performance (Jones et al. 2010; Morton 2006).

The parameters of the power-duration relationship are conventionally determined by modeling the relationship between work rate and time to exhaustion (T_{lim}) as derived from a series of constant work rate (CWR) exercise tests (Hill 1993; Housh et al. 1989; Jones et al. 2008a; Moritani et al. 1981; Poole et al. 1988). It has been suggested, however, that time trial (TT) tests, where subjects manipulate their own pace to complete a set distance or amount of work as quickly as possible, are more reliable than CWR tests and more closely replicate competitive sport (Hopkins et al. 2001; Jeukendrup et al. 1996). While TT-type performance tests have been used in the field to determine the power-duration relationship (Karsten et al. 2014; Quod et al. 2010), it is presently not known whether CWR and TT protocols performed under identical experimental conditions yield equivalent CP and W′ parameter estimates.

It has been suggested that a self-selected 'parabolic' pacing strategy, typically involving a fast start, may enable the individual to maintain a higher mean power output for the duration of a TT than would be possible with an externally-imposed CWR test (Thomas et al. 2013). A fast start pacing strategy has been shown to augment the $\dot{V}O_2$ response early in exercise and lead to a greater tolerable duration of severe-intensity exercise than would be predicted based on the CP and W′ derived from CWR prediction trials (Jones et al. 2008b). These findings suggest that a fast start pacing strategy may improve exercise performance by increasing CP, W′ or both. If the power-duration relationship is significantly impacted by variations in pacing strategy, this may have considerable implications for the accuracy with which competitive TT performance can be predicted based on the CP and W′ as traditionally measured. There is a strong body of evidence to suggest that adopting a fast start pacing strategy increases the rate at which $\dot{V}O_2$ increases towards its maximum and improves severe-intensity exercise performance compared to a CWR pacing strategy (Aisbett et al. 2009; Bailey et al. 2011; Hettinga et al. 2009; Jones et al. 2008b). As the CP has been shown to be inversely correlated with the phase II $\dot{V}O_2$ time constant (τ) (Murgatroyd et al. 2011), it is conceivable that a parabolic pacing strategy, that is typically self-selected during a TT, may augment the $\dot{V}O_2$ response at exercise onset and increase the CP.

The purpose of this study, therefore, was to determine the effect of pacing strategy during severe-intensity prediction trials on the CP and the $\dot{V}O_2$ mean response time (MRT). We tested the hypotheses that the CP derived from a series of self-paced TTs would be greater than the CP derived from CWR prediction trials, and that the $\dot{V}O_2$MRT would be lower during TT compared to CWR prediction trials. A secondary purpose of this study was to evaluate the accuracy of time-trial performance prediction on the basis of power-duration parameters derived from a CWR prediction trial protocol.

METHODS

Subjects

Ten healthy, physically active male subjects (mean ± SD: age 21.5 ± 1.9 years, height 1.79 ± 0.07 m, mass 75.2 ± 11.5 kg) volunteered to participate in this study. The study was approved by the University of Exeter Research Ethics Committee. Written informed consent was obtained and a physical activity readiness questionnaire (PAR-Q) was completed prior to data collection. Subjects were instructed to report to all testing sessions well-hydrated, having avoided strenuous physical activity and caffeine ingestion for 24 h and 3 h prior to testing, respectively. Testing was completed at the same time of day (± 2 h) for each subject and laboratory visits were separated by at least 24 h.

Experimental design

Subjects reported to the laboratory on 11-18 occasions over a 6-wk period, such that subjects typically completed 2-3 exercise tests per week. All tests were performed on an electronically-braked cycle ergometer (Lode, Excalibur, Groningen, The Netherlands). The ergometer seat and handlebar height were adjusted for comfort and the same settings were replicated for subsequent tests. Subjects performed a ramp incremental test to volitional exhaustion for the determination of the gas exchange threshold (GET) and the peak O_2 uptake ($\dot{V}O_2max$). Subjects then performed 3-4 CWR prediction trials and 3-4 TT prediction trials in a semi-randomised order for the determination of the power-duration relationship. The CWR trial always preceded the corresponding TT to enable the calculation of an appropriate fixed resistance (i.e., linear factor) for the TT and to ensure similar durations and cadences of CWR and TT prediction trials. Trials were randomised such that the comparative TT test was not always administered immediately after the comparative CWR test (by way of example, test 1: CWR3, test 2: CWR2, test 3: TT3, test 4: TT2, test 5: CWR1, test 6: TT1 etc.). Each TT prediction trial was preceded by at least one familiarisation session. Following the completion of all CWR and TT prediction trials, the ramp incremental test was repeated to assess potential training effects that might have occurred during the study period.

Determination of the GET and $\dot{V}O_2$peak

The ramp incremental test included 4 min of 'unloaded' (20 W) pedaling, followed by a ramp increase in power of 30 W·min^{-1}. Subjects were instructed to maintain their preferred cadence (80 rpm, *n* = 7; 90 rpm, *n* = 3) throughout the test. The test was terminated when the cadence fell by more than 10 rpm below the preferred cadence for more than 5 s despite strong verbal encouragement. The GET was determined as: 1) the first disproportionate increase in carbon dioxide output ($\dot{V}CO_2$) versus $\dot{V}O_2$; 2) an increase in minute ventilation ($\dot{V}_E$) relative to $\dot{V}O_2$ with no increase in $\dot{V}_E$/$\dot{V}CO_2$; and 3) the first increase in end-tidal O_2 tension with no fall in end-tidal CO_2 tension. The GET was estimated independently by three experienced assessors using code-labeled data files. In all cases at least two assessors identified the same value. The $\dot{V}O_2$peak was determined as the highest 15-s rolling mean $\dot{V}O_2$ recorded during the test.

Determination of the power-duration relationship using CWR trials

CP and W′ were estimated from 3-4 CWR prediction trials (3 trials, n = 5; 4 trials, n = 5), performed at different work rates (approximately 60%Δ, 70% Δ, 80% Δ and 100% $\dot{V}O_2$peak; where Δ refers to the work rate difference between the GET and the $\dot{V}O_2$peak). As a quality control measure of the mathematical modeling of power-duration parameters, *a priori* criteria were set for the standard errors associated with the CP and W′, such that, if the standard errors associated with the CP and W′ exceeded 5 and 10 %, respectively, after three prediction trials had been performed, a fourth prediction trial was completed. Any prediction trials where the end-exercise $\dot{V}O_2$was <95% of the individual's ramp test determined $\dot{V}O_2$peak were excluded from the modeling of the power-duration relationship. The work rates for each trial were selected to obtain a range of T_{lim} between approximately 2 and 15 min, with a minimum of 5 min between the shortest and longest trials. Each trial started with 4 min of unloaded pedaling at the subject's preferred cadence, followed by a step increase to the required work rate. Subjects were instructed to remain seated and to maintain their preferred cadence for as long as possible. Strong verbal encouragement was provided throughout, but subjects were not informed of the work rate or the elapsed time. The test was terminated when cadence fell by more than 10 rpm below the preferred cadence for more than 5 s. The T_{lim} was recorded to the nearest second.

Determination of the power-duration relationship using TTs

The total work done during each CWR prediction trial for each subject was used as a 'target total work done' for the corresponding self-paced, maximal TT. The Lode Excalibur Sport is an electro-magnetically braked ergometer, which enables time trials to be performed against a fixed resistance. For the TTs, the ergometer was set on the so-called 'linear mode', where the fixed resistance (i.e. linear factor) is set according to the equation Linear factor= CWR/rpm^2, where CWR represents the work rate in the given CWR prediction trial and rpm represents the cadence self-selected by the subject during the initial ramp test. The same *a priori* criteria for the standard errors were applied as for the CWR protocol, such that if the standard errors associated with the CP and W′ exceeded 5 and 10%, respectively, after three prediction trials had been performed, a fourth prediction trial was completed. Had a subject performed only 3 CWR prediction trials, the amount of work to be completed during the fourth TT was estimated using *Equation 2*, and the power output for the determination of the linear factor was estimated using *Equation 3*. Therefore, the CP and W′ were estimated from 3-4 TT prediction trials (3 trials, n = 3; 4 trials, n = 7), and the work rates for each trial were selected to obtain a range of T_{lim} between approximately 2 and 15 min, with a minimum of 5 min between the shortest and longest trials. Each test started with 4 min of unloaded pedaling at each subject's preferred cadence, followed immediately by the TT.

The linear mode does not allow the subject to change the linear factor (i.e. the 'gear') during the protocol. In other words, the conditions during the TT simulated a situation where the subject pedals a bicycle along a level road, and while it is not possible to change the gear, speed can be varied by varying cadence. For maximal TT performance, subjects were instructed to complete the given 'target work done' as quickly as possible. Subjects were familiarized to each self-paced 'target work done', such that they performed a minimum of two trials at each 'target work done' where the criteria for a maximal TT performance were as follows: 1) attainment of >95% of the $\dot{V}O_2$peak measured in the ramp incremental test; and 2) the performance times between the two consecutive TTs varying by less than 5%. Upon meeting these criteria, the shortest time taken to achieve the designated amount of work for each TT was used for analysis. Visual and verbal feedback regarding the amount of work completed was deliberately provided throughout the trials to mimic real-world TT conditions.

Subjects were instructed to remain seated and were provided with strong verbal encouragement.

In all tests, subjects wore a nose clip and breathed through a low-dead space (90 mL), low resistance (0.75 mmHg.L^{-1}.s^{-1}) mouthpiece and impeller turbine transducer assembly (Jaeger Triple V, Jaeger GmbH, Hoechburg, Germany). Inspired and expired gas volume and concentration signals were continuously sampled at 100 Hz. The gases were drawn continuously from the mouthpiece through a 1.5 m sampling line (0.5 mm internal diameter) to paramagnetic (O_2) and infrared (CO_2) analyzers (Jaeger Oxycon Pro, Hoechburg, Germany). These analyzers were calibrated before each test with gases of known concentration, and the turbine volume transducer was calibrated using a 3-L syringe (Hans Rudolph, KS). The volume and concentration signals were time-aligned, accounting for the transit delay in capillary gas and analyser rise time relative to volume signal. The $\dot{V}O_2$, $\dot{V}CO_2$ and $\dot{V}_E$ were calculated for each breath using standard formulae. The peak $\dot{V}O_2$during the CWR and TT prediction trials was calculated as the highest 15-s rolling mean value.

Data Analyses

The CP and W′ parameters were estimated using three models: the hyperbolic P-T_{lim} model (Equation 1); the linear work-time (W-T_{lim}) model, where the total work done (W) is plotted against time (Equation 2); and the linear inverse-of-time (1/T_{lim}) model, where power output is plotted against the inverse of time (Equation 3):

$$T_{lim} = W'/(P - CP) \qquad [1]$$

$$W = CP \cdot T_{lim} + W' \qquad [2]$$

$$P = W' \cdot \left(\frac{1}{T_{lim}}\right) + CP \qquad [3]$$

The power (P) during the TTs was defined as the mean power output measured across the duration of each trial. The SEE associated with the CP and W′ were expressed as coefficients of variation (CV%, i.e. relative to the parameter estimate). The 'total error' associated with the modeling of the power-duration parameters was calculated as the sum of the CV% associated with the CP and the CV% associated with the W′. The sum of the CV% was optimised for each individual by selecting the model with the smallest total error (Equation 1, 2 or 3) to produce the 'best individual fit' parameter estimates. Similarly, the parameter estimates from a model associated with the largest total error were grouped together to produce the 'worst individual fit' parameter estimates. The 'best fit' and 'worst fit' CP and W′ derived from the CWR prediction trials were then used to calculate the predicted TT completion time (T_{lim}) for each 'target work done' (W) by re-arranging Equation 2:

$$T_{lim} = (W - W')/CP \qquad [4]$$

The breath-by-breath $\dot{V}O_2$data from each test were initially examined to exclude errant breaths caused by coughing, swallowing, etc., and those values lying more than 4 SD from the local mean were removed. Subsequently, the breath-by-breath data were converted to second-by-second data using linear interpolation and time aligned to the start of the exercise. A nonlinear least squares algorithm was used to fit the data. Given that the subjects only completed one trial at each work rate, it was not justified to use a bi-exponential model to characterise the $\dot{V}O_2$ kinetics as the statistical confidence in the derived parameter estimates would be low. Therefore, a single exponential model without time delay, with the fitting window commencing at t = 0 s (equivalent to the mean response time, MRT), was used to characterise the overall $\dot{V}O_2$ response as described in the following equation:

$$\dot{V}O_2(t) = \dot{V}O_{2\,baseline} + A \cdot (1 - e^{-\frac{t}{\tau}}) \quad [5]$$

where $\dot{V}O_2$ (t) represents the absolute $\dot{V}O_2$at a given time t, $\dot{V}O_2$baseline is the average of the $\dot{V}O_2$measured over the final 120 s of baseline pedaling, A and τ represent the amplitude and time constant, respectively, describing the overall increase in $\dot{V}O_2$above baseline. The total O_2 consumed (in litres) was also calculated for the first 60 s of each exercise bout as well as for the entire exercise bout.

Statistical Analyses

Two-way ANOVAs with repeated measures were conducted to assess differences in power-duration parameters between models (Equations 1-3 and the best and worst individual fits) and protocols (CWR vs. TT). Independent samples one-way ANOVA was used to assess differences in the peak $\dot{V}O_2$achieved in the TT (n=37) and CWR (n=35) tests and the ramp test $\dot{V}O_2$peak (n=10 pre and post). Paired samples *t*-tests were used to evaluate differences in the $\dot{V}O_2$MRT, total O_2 consumed, mean power output, the mean power output over the first 30 s of the TT, and trial completion time between the corresponding work-matched CWR and TT prediction trials (n=33). Paired samples *t*-tests were also used to evaluate differences in the actual vs. predicted TT completion times (n=37) between the CWR best and worst individual fits. Pearson's product moment correlation coefficient was used to assess the relationships between the $\dot{V}O_2$ MRT and the CP (n=20), the total O_2 consumed and the CP (n=20), the relationship between the relative change in CP and the relative change in W′

between the CWR and TT protocols (n=10), and the relationship between actual and predicted TT performance (n=37). Statistical significance was accepted at P<0.05 and data are presented as mean ± SD.

RESULTS

The $\dot{V}O_2$peak measured in the initial ramp incremental test was 4.14 ± 0.56 L·min^{-1} (56 ± 6 mL·kg^{-1}·min^{-1}) and the peak power output was 377 ± 59 W. The GET occurred at 2.06 ± 0.33 L·min^{-1} and 122 ± 30 W. There was no significant difference between the pre- and post-study $\dot{V}O_2$peak determined during ramp incremental exercise (P>0.05).

The power-duration parameters

The T_{lim} in the CWR prediction trials was 384 ±189 s (range: 135 to 896 s). The $\dot{V}O_2$peak values in the CWR prediction trials (4.14 ± 0.47 L·min^{-1}) were not significantly different from the $\dot{V}O_2$peak achieved during the ramp incremental test (P>0.05). The mean cadence during the CWR prediction trials was 83 ± 4 rpm. There were no differences in CP or W′ estimates between the three models (Equations 1-3), or the ‘best individual fit’ and the ‘worst individual fit’ within the CWR protocol (P>0.05; Table 1). The CP estimate from the ‘best individual fit’ corresponded to 66 ± 4% of ramp incremental test peak power and 46 ± 8%Δ.

The completion times of the TT prediction trials was 371 ± 185 s (range: 135 to 899 s). The TT completion times were significantly shorter than the T_{lim} in the respective CWR prediction trials (P<0.05). The $\dot{V}O_2$peak values in the TT prediction trials (4.21 ± 0.45 L·min^{-1}) were not significantly different from the values attained in the CWR prediction trials, or the values achieved during the ramp incremental test (P>0.05). The mean cadence during the TT prediction trials was 84 ± 5 rpm. There were no differences in CP or W′ estimates between the three models (Equations 1-3), or the 'best individual fit' and the worst individual fit' within the TT protocol (*P*>0.05; Table 1). The CP estimate from the 'best individual fit' corresponded to 71 ± 8% ramp incremental test peak power and 53 ± 12%Δ.

The mean power outputs maintained during corresponding work-matched prediction trials (*n*=33) were significantly greater during the TT prediction trials (330 ± 56 W) compared to the CWR prediction trials (318 ± 59 W) (*P*<0.05). The mean power output over the first 30 s of the TT tests (357 ± 78 W) was significantly greater than the mean power output during the CWR prediction trials (*P*<0.05; n=33), representing 107 ± 8% of the TT mean power. The pacing strategy of a representative subject is displayed in Figure 1.

The 'best individual fit' for the CWR protocol was derived from the P-T_{lim} model in 1 subject, W-T_{lim} model in 4 subjects and 1/T_{lim} model in 5 subjects, whereas the 'best individual fit' for the TT protocol was derived from the W-T_{lim} model in 1 subject and 1/T_{lim} model in 9 subjects. There were no differences (*P*>0.05) in the standard errors (SEE) associated with the CP or W′ modeling between the CWR and TT protocols (Table 1). However, there were significant differences in the SEEs associated with the W′ parameter and the 'total error' between models *within* the CWR and TT protocols (specific differences indicated in Table 1).

As there were no differences in the power-duration parameters derived from different models within the CWR and TT protocols (Table 1), the estimates from the 'best individual fit', which were associated with the lowest errors, were chosen for further analyses. The CP derived from the TT protocol was ~7% greater than that derived from the CWR protocol

(P<0.05; Figure 2A). The CP estimates from the TT and CWR protocols were positively correlated (r=0.91, P<0.01). The W′ estimates were not significantly different between the CWR and TT protocols (P>0.05; Figure 2B). The W′ estimates from the two protocols were correlated (r=0.67, P<0.05). There was a significant inverse correlation between the relative difference in CP (ΔCP) and the relative difference in W′ (ΔW′) between the CWR and TT protocols (r=-0.74, P<0.01).

$\dot{V}O_2$ response

The overall rate of increase in $\dot{V}O_2$ during the prediction trials was characterised using the MRT. The MRT during the corresponding work-matched CWR and TT prediction trials (n=33) was significantly shorter during the TT prediction trials (34 ± 16 s) compared to the CWR prediction trials (39 ± 19 s, P<0.05; Figure 2C). The mean MRT for all CWR prediction trials for each individual, and the mean MRT for all TT prediction trials for each individual, were calculated. When combined, these CWR and TT protocol mean MRTs (n=20) were inversely correlated with the CP derived from the respective protocols (r=-0.66, P<0.05).

The total O_2 consumed during the first 60 s of exercise was significantly greater during the corresponding work-matched TT (2.87 ± 0.43 L) compared to the CWR (2.75 ± 0.45 L) prediction trials (P<0.05). The total work done during the first 60 s was also significantly greater during the corresponding work matched TT (21.1 ± 4.5 kJ) compared to the CWR (19.3 ± 3.6 kJ) prediction trials (P<0.05), such that the total work done per unit O_2 consumed over the first 60 s was slightly but significantly greater during the TTs (7.3 ± 0.7 $kJ \cdot L^{-1} \cdot min^{-1}$) than the CWR trials (7.0 ± 0.6 $kJ \cdot L^{-1} \cdot min^{-1}$, P<0.05). The mean of total O_2 consumed in the first 60 s for all CWR trials for each individual, and the mean of total O_2 consumed in the first 60 s for all TT trials for each individual were calculated. When combined, these CWR and TT protocol means for O_2 consumed in the first 60 s (n=20) were positively correlated with the CP derived from the respective protocols (r=0.88, P<0.05). The total O_2 consumed during the entire exercise bout was not different between the work-matched CWR (21.44 ± 10.33 L) and TT (21.36 ± 10.04 L) prediction trials (P>0.05; n=33).

There was no significant difference between the actual TT completion time (356 ± 185 s) and the predicted completion time calculated according to *Equation 4* and the 'best individual fit' CP and W′ estimates from the CWR protocol (362 ± 189 s)(P=0.11; n=37). However, the predicted TT completion time based on the 'worst individual fit' from the CWR protocol (451 ± 188 s) was significantly greater than the actual completion time (P<0.05; n=37).

DISCUSSION

The principal novel findings of this study were that: 1) the CP derived from a series of TT prediction trials was significantly greater compared to that derived from a series of CWR prediction trials; 2) the W′ was not significantly affected by differences in pacing strategy (TT vs. CWR); and, 3) the $\dot{V}O_2$MRT was significantly greater during the CWR compared to the TT tests, with a significant inverse relationship between the MRT and the CP. These results indicate that performance was improved in the TT compared to CWR tests, concomitant with a greater CP and augmented $\dot{V}O_2$response, whereas W′ was not significantly influenced by differences in pacing between the CWR and TT prediction trials. Direct quantification of the effects of pacing on the power-duration parameters is a key novel contribution of this study and, collectively, the present findings offer important novel insights into the potential mechanisms by which a fast starting strategy may be ergogenic. Our findings also highlight a potential source of error, stemming from a significant difference in CP between protocols, when predicting TT performance on the basis of CWR prediction trials.

The CP estimate was ~7% greater when derived from the TT protocol compared to the CWR protocol. The magnitude of this increase is similar to the influence of cadence on CP (~6% difference between 60 rpm vs. 100 rpm; Barker et al. 2006; Hill et al. 1995) but smaller than the increase in CP observed following high-intensity interval training (~10-15%; Gaesser & Wilson, 1988, Vanhatalo et al. 2008). The power profiles of the self-paced TT prediction trials were indicative of a 'parabolic' pacing strategy where the power outputs over the early stages of the trial were higher than the mean power output, followed by a gradual decline in power and (with the exception of the shortest trial) a subsequent increase in power during a 'sprint finish'. The shorter $\dot{V}O_2$MRT and greater total O_2 consumed during the first 60 s in the TT protocol were likely to result from higher power outputs at the onset of the TT compared to the CWR trials, which would generate a greater 'error signal' between the muscle ATP requirement and the rate of oxidative phosphorylation in the early phase of exercise (Bailey et al. 2011, Jones et al. 2008b). It is possible that the greater rate of increase in oxidative phosphorylation during the self-paced TT relative to the CWR trials reduced the reliance on non-oxidative metabolism (i.e. the O_2 deficit) across the exercise transient (but see caveats discussed later), thereby reducing the extent of metabolic perturbation early on in exercise and enabling a greater power output to be maintained throughout the trial (Bailey et al. 2011; Jones et al. 2008b). The improved exercise performance associated with the fast-start pacing strategy during the self-paced TTs compared to CWR tests is compatible with this interpretation and was reflected in the power-duration relationship as an elevated CP with no significant change in the W′.

The present study showed no statistically significant change in the size of the available W′ despite the different rates of W′ utilisation between CWR and self-paced TT protocols. However, the change in W′ was inversely correlated with the increase in CP, such that the individuals with the greatest increases in CP had the greatest reductions in the W′. The inter-related nature of the W′ and CP has been previously illustrated following interventions such as training and inspiration of hyperoxic gas (Jenkins and Quigley 1992; Vanhatalo et al. 2010), which are known to reduce the amplitude of the $\dot{V}O_2$ slow component (Jones et al. 2011). It has been proposed that the inter-individual variability in the changes in W′ following interventions which influence the $\dot{V}O_2$kinetics may depend on the relative changes in the CP and the $\dot{V}O_2$peak (Bailey et al. 2011; Jones et al. 2010; Vanhatalo et al. 2010). In the present study, the $\dot{V}O_2$ peak remained unaltered but the CP increased in the TT compared to the CWR protocol, indicating that the ‘range’ of the severe domain (defined as the relative difference between CP and $\dot{V}O_2$ peak), and therefore the scope for the $\dot{V}O_2$ slow component to develop, was smaller in the TT protocol. It may be speculated that the significant inverse correlation between the changes in W′ and CP may reflect a smaller $\dot{V}O_2$ slow component, consequent to fast-start pacing strategy, during TT compared to CWR trials.

The fast-start pacing strategy adopted by the subjects during the self-paced TTs significantly augmented the $\dot{V}O_2$ response following exercise onset compared to the CWR tests (Figure 2C). This finding is consistent with previous studies that have reported a greater rate of $\dot{V}O_2$ increase when a fast-start strategy is adopted (Aisbett et al. 2009; Bailey et al. 2011; Bishop et al. 2002; Hanon et al. 2008; Jones et al. 2008b). However, not all reports support the performance enhancing effect of fast-start pacing strategy (Hanon et al. 2008; Mattern et al. 2001). Bailey et al. (2011) showed that an externally imposed fast-start pacing strategy increased the rate at which $\dot{V}O_2$ increased towards its maximum during both 3-min and 6-min trial durations, but exercise performance was only improved in the 3-min performance trial. This difference was attributed, in part, to the observation that the subjects were only able to attain $\dot{V}O_2$ peak in the 3-min trial when using a fast-start, but not an even-pace or slow-start, pacing strategy (Bailey et al. 2011). In the present study, the $\dot{V}O_2$ peak was consistently attained in all TT and CWR prediction trials, suggesting that the underlying mechanism for the inferior performance during the CWR compared to TT tests was not associated with the inability to attain $\dot{V}O_2$ peak and/or the incomplete utilisation of the W′ (indeed, the W′ was not significantly influenced by self-pacing). It is possible that the different durations of the fast-start phase in the study by Bailey et al. (2011) resulted in a greater demand on the anaerobic metabolic pathways during the 6-min than the 3-min trials, with a concomitant increase in the O_2 deficit. In the present study, subjects were allowed to self-select their pacing strategy as opposed to having the fast-start imposed externally (Bailey et al. 2011). This may have enabled subjects to preserve a sufficient proportion of the W′, thus permitting a greater mean power output to be achieved throughout the TT compared to the CWR tests. In this regard, the potential ergogenic effect of a fast-start strategy on performance appears to be dependent on the dynamic relationship between event duration, and the duration and intensity of the fast-start phase (Bailey et al. 2011; Foster et al. 2004; Wood et al. 2014).

The tolerable duration of severe-intensity exercise has been proposed to be determined by the characteristics of the $\dot{V}O_2$ kinetic response, the $\dot{V}O_2$ peak and the ‘anaerobic capacity’ (Burnley and Jones 2007). These variables are associated with the parameters that define the power-duration relationship: the CP has been shown to be related to the rate at which the

$\dot{V}O_2$ increases at exercise onset (i.e. the phase II τ; Murgatroyd et al. 2011) and the W′ is related to the amplitude of the $\dot{V}O_2$ slow component (Murgatroyd et al. 2011; Vanhatalo et al. 2011). Given the breath-by-breath variability inherent in pulmonary $\dot{V}O_2$measurement, accurate characterization of the $\dot{V}O_2$ phase II τ requires the averaging of $\dot{V}O_2$ responses from several step transitions in work rate. Since only a single baseline-to-exercise transition was completed for each prediction trial in the present study, we determined the MRT of the $\dot{V}O_2$ response to provide a broad assessment of the rate at which $\dot{V}O_2$ projected towards its maximum.

It is important to note that ATP turnover rate is not constant during TT exercise and the shorter MRT in the TT compared to CWR tests should not be considered indicative of faster $\dot{V}O_2$ kinetics *per se*. Nevertheless, the MRT assessment in this study enabled comparison of the rate at which $\dot{V}O_2$ projected towards its maximum value between the TT and CWR protocols. The significant relationships between CP and the MRT (r=-0.66), and the CP and the total O_2 consumed during the first 60 s of exercise (r=0.88), may suggest that the rate of $\dot{V}O_2$ increase during the exercise bout influences the power-duration relationship and severe-intensity exercise performance. Our finding of an inverse correlation between the CP and the $\dot{V}O_2$ MRT is consistent with the inverse relationship between CP and the $\dot{V}O_2$ phase II τ reported previously (Murgatroyd et al. 2011). The findings of the current study, therefore, support the notion that $\dot{V}O_2$ kinetics and the parameters of the power-duration relationship may share similar physiological determinants (Burnley and Jones 2007; Jones et al. 2010; Murgatroyd et al. 2011; Vanhatalo et al. 2011).

The augmented $\dot{V}O_2$ early in the TTs compared to the CWR trials is a direct reflection of the fast-start strategy employed in the former, and therefore the higher power outputs selected. The fast-start pacing strategy observed during the TT led to enhanced CP (with no significant changes in $\dot{V}O_2$ peak and W′) and improved severe-intensity exercise performance. It appears that the fast-start pacing strategy selected by subjects does not accelerate substrate-level phosphorylation to the extent that it invokes a sufficient muscle metabolic perturbation to precipitate fatigue (and result in premature depletion of the W′). While ATP turnover is not constant over time during TT (or even CWR) exercise (Bangsbo et al. 2001), there is evidence that muscle efficiency may be higher in the early compared to the later stages of an exercise transient (Krustrup et al. 2003; Wüst et al. 2011). It may be speculated that subjects can take advantage of this metabolic 'window of opportunity' wherein the selection of relatively high initial power outputs enables a more rapid acceleration of $\dot{V}O_2$ without invoking a substantial, and deleterious, metabolic cost.

It is assumed that when the CP and W′ are known, the power-duration relationship (Equations 1-3) can be used to predict performance in any continuous exercise bout provided that the work rate remains within the severe intensity exercise domain (Hill 1993; Jones et al. 2010; Morton 2006). We used Equation 4 and the CP and W′ estimates derived from the CWR prediction trials to predict TT performance. The predicted TT duration was overestimated (on average by ~27%) when 'worst individual fit' parameters were used and tended to be overestimated (by ~2%) even when the 'best individual fit' parameters were used due to the significantly greater CP during TT compared to CWR exercise (Figure 3). It should be noted that the coefficient of variation between actual and predicted TT performance was ~6%, which is slightly greater than the typical test-retest reliability of TT performance of similar durations (0.6 - 4.6%; Hopkins et al. 2001). For comparison, the accuracy of TT prediction in the present study was similar to the accuracy of predicting T_{lim} (~2-13%) during CWR exercise on the basis of the power-duration relationship derived from CWR prediction trials (Housh et al. 1989). Although it may be argued that the magnitude of error in TT performance prediction on the basis of CWR prediction trials is only marginally greater than the reliability of TT performance, the significant difference in CP between CWR and TT exercise nevertheless results in a consistent under-prediction of TT performance which is likely to be meaningful when attempting to predict marginal alterations in athletic performance. It is recommended that the 'best individual fit' CP and W′ estimates are established using equations 1-3 to minimize the potential prediction error arising from differences in pacing.

The close agreement between the parameter estimates derived from equations 1-3 within the CWR and TT protocols indicates that the incidence of random and systematic errors in the prediction trial data was low (Hill and Smith 1994). The goodness of fit was further demonstrated by the finding that the error associated with the modeled CP was not significantly different between the best (~1-2%) and worst (~3-4%) individual fits (Table 1). However, the error associated with the W′ was significantly greater for the worst (~19-21%) compared to the best (~7-9%) individual fit, which likely negatively influenced the accuracy of performance prediction (Figure 3). It is important to note that if inconsistencies in the attainment of $\dot{V}O_2$ peak in the prediction trials occur, this may impact on the modelled power-duration parameters. Inclusion of data from prediction trials where the $\dot{V}O_2$ peak has not been attained has been associated with an underestimation of CP (Sawyer et al. 2012). The criteria of: 1) no significant differences in parameter estimates between models (equations 1-3); and, 2) attainment of >95% of $\dot{V}O_2$ peak in all CWR prediction trials are therefore advisable to ensure reasonably accurate TT performance prediction (CV ~5-9%) when the ‘best individual fit’ parameter estimates are used.

In addition to the physiological determinants of performance, which were the main focus of the present study, it should be noted that subject motivation also plays a role in determining maximal high-intensity exercise performance. The participants in this study were instructed and strongly encouraged to provide a maximal effort in all tests (CWR and TT) and were challenged to achieve their $\dot{V}O_2$ peak every time. Indeed, attainment of $\dot{V}O_2$ peak was used as a criterion of maximal effort, indicating consistent metabolic strain across all trials. Although verbal encouragement was kept consistent between conditions, it is possible that subject motivation might have been influenced by the feedback given on work completed during the TTs but not during the CWR tests. Potential differences in subject motivation between time-to-exhaustion, open-ended CWR tests and the fixed-endpoint TTs might have contributed to differences in performance to some extent. The definitive assessment of the potential influence of motivation on the power-duration relationship is beyond the scope of the present study and warrants further investigation.

Recent development of user-friendly on-bike power meters now affords an opportunity to assess maximal performance within the power-duration framework in an actual competitive situation on the road. An additional advantage of on-bike power meters is that the cyclist is able to control the gear as well as the cadence during performance, as opposed to the ‘fixed resistance - variable cadence’ model applied in the present study. The aim of the current study was to investigate in a controlled laboratory environment whether prediction trial protocols with distinct power profiles (Figure 1) result in similar CP and W′ estimates. Our finding that the CP was significantly greater in a TT than CWR protocol opens up interesting avenues for field assessment of power output distribution during competitive TTs.

In conclusion, the CP was greater and the $\dot{V}O_2$ MRT was shorter during self-paced TT exercise compared to CWR exercise, while the W′ was not significantly affected by differences in pacing strategy. The significant difference in CP between the TT and CWR protocols was reflected in a small but consistent under-prediction of TT performance when the CP and W′ were derived from CWR prediction trials. It is therefore recommended that the 'best individual fit' CP and W′ estimates are established using equations 1-3, and where possible, TT prediction protocols, in which subjects are permitted to self-pace, are used for determination of the CP and W′ when accurate predictions of performance are desired. This approach may represent an important step forward in the application of the power-duration relationship to modeling athletic performance.

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Table 1 The parameter estimates derived from Equations 1-3 and the best and worst individual fits for the TT and CWR protocols. Total error indicates the sum of CV% associated with CP and W′.

		R^2	CP (W)	SEE (W)	CV%	W′ (kJ)	SEE (kJ)	CV%	Total error (CV%)
CWR protocol	**W-T_{lim} model**	0.996 – 1.000	248 ± 44	4.6 ± 5.6	1.7 ± 1.7	21.4 ± 7.0	1.8 ± 2.0	9.5 ± 9.8	11.2 ± 11.5
	1/T_{lim} model	0.964 – 0.998	251 ± 47	4.9 ± 3.1	1.9 ± 1.0	19.1 ± 5.1	1.3 ± 0.7	7.3 ± 4.7	9.3 ± 5.6 [c]
	P-T_{lim} model	0.965 – 0.999	247 ± 43	4.8 ± 4.7	1.9 ± 1.6	20.6 ± 6.0	2.5 ± 2.1 [c]	12.4 ± 9.3 [c e]	14.3 ± 10.8 [c]
	Best individual fit	0.974 – 1.000	250 ± 47	3.6 ± 3.3	1.4 ± 0.9	20.6 ± 7.4	1.2 ± 0.7 [d]	6.7 ± 4.9 [d]	8.0 ± 5.7 [d]
	Worst individual fit	0.987 – 0.998	247 ± 44	10.0 ± 6.6	4.1 ± 2.7	20.3 ± 5.2	3.6 ± 2.2 [c]	18.5 ± 10.7 [c]	22.6 ± 12.4 [c]
TT protocol	**W-T_{lim} model**	0.996 – 1.000	265 ± 45 [a]	5.3 ± 2.3	2.1 ± 1.2	18.5 ± 6.8	2.1 ± 0.9 [c e]	12.3 ± 5.1 [c e]	14.4 ± 5.7 [c e]
	1/T_{lim} model	0.935 – 0.999	265 ± 44 [a]	5.9 ± 3.0	2.3 ± 1.3	18.0 ± 5.7	1.4 ± 0.8 [d]	8.9 ± 5.2 [d]	11.2 ± 6.2 [d]
	P-T_{lim} model	0.804 – 0.994	264 ± 45 [b]	7.1 ± 6.2	2.6 ± 2.0	18.4 ± 8.4	3.4 ± 2.5 [c e]	20.7 ± 13.1 [c e]	23.3 ± 14.5 [c e]
	Best individual fit	0.983 – 1.000	265 ± 44 [a]	5.5 ± 2.8	2.2 ± 1.3	18.1 ± 5.7	1.4 ± 0.8 [d]	8.5 ± 4.9 [d]	10.7 ± 6.0 [d]
	Worst individual fit	0.804 – 0.998	265 ± 44 [b]	7.5 ± 6.0	2.8 ± 1.9	17.8 ± 7.3	3.4 ± 2.5 [c]	20.8 ± 13.0 [c]	23.6 ± 14.3 [c]

a Significantly different from the respective CWR protocol (P<0.05). **b** Statistical trend towards difference from the respective CWR protocol (P<0.06). **c** Significantly different from the 'Best individual fit', **d** significantly different from the 'Worst individual fit' and **e** significantly different from the '1/T_{lim} model' within the same protocol (P<0.05).

FIGURE LEGENDS

Figure 1: The pacing strategy of a representative subject during short (panel A), short-intermediate (panel B), long-intermediate (panel C), and long (panel D) TTs. The power output of the corresponding CWR prediction trial is shown as a dashed line in each panel.

Figure 2: The group mean (±SD) CP derived from the TT protocol was 6% greater than the CP derived from the CWR protocol (panel A). The individual differences in CP between TT and CWR protocols ranged from -2 to +57 W. The group mean W′ was 12% lower in TT compared to CWR protocol (panel B) and the individual differences ranged from -14.8 to +3.4 kJ. In panels A and B, the solid lines indicate individual responses. The group mean MRT derived from all TT tests was 14% shorter compared to the CWR protocol (panel C). Individual responses (n=33) are not shown in panel C for clarity. **P*<0.05.

Figure 3: The relationships between actual TT performance and TT performance predicted on the basis of the 'best' and 'worst' individual fits derived from the CWR trials (panels A and B, respectively). The predicted TT durations derived from the 'best individual fit' were not significantly different from the actual TT durations whereas the TT durations predicted from the 'worst individual fit' were significantly different from the actual TT durations. **P*<0.05.

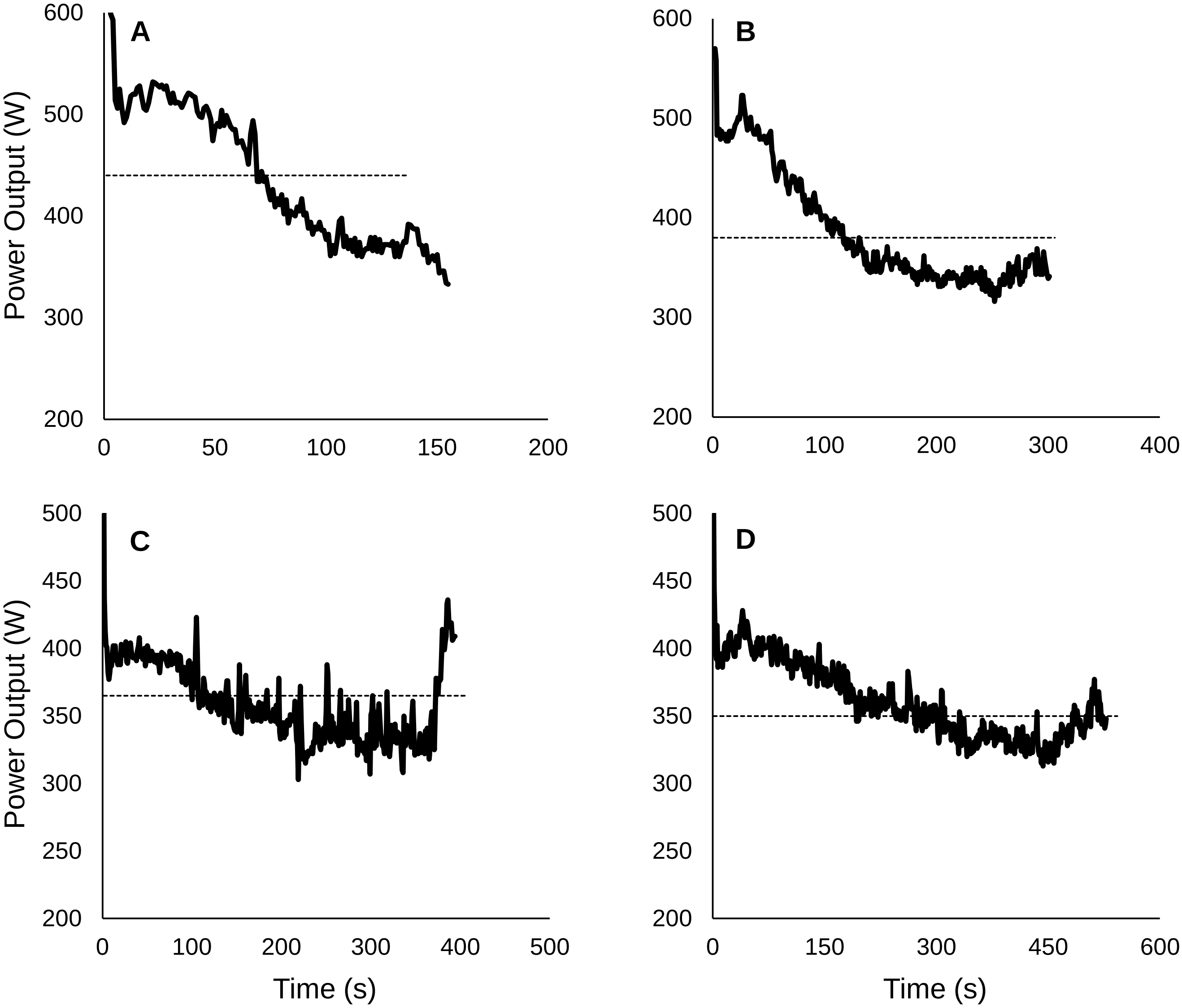
A
B
C
D
Power Output (W)
Power Output (W)
Time (s)
Time (s)
600
500
400
300
200
0
50
100
150
200
0
100
200
300
400
500
450
350
250
600

Figure 1

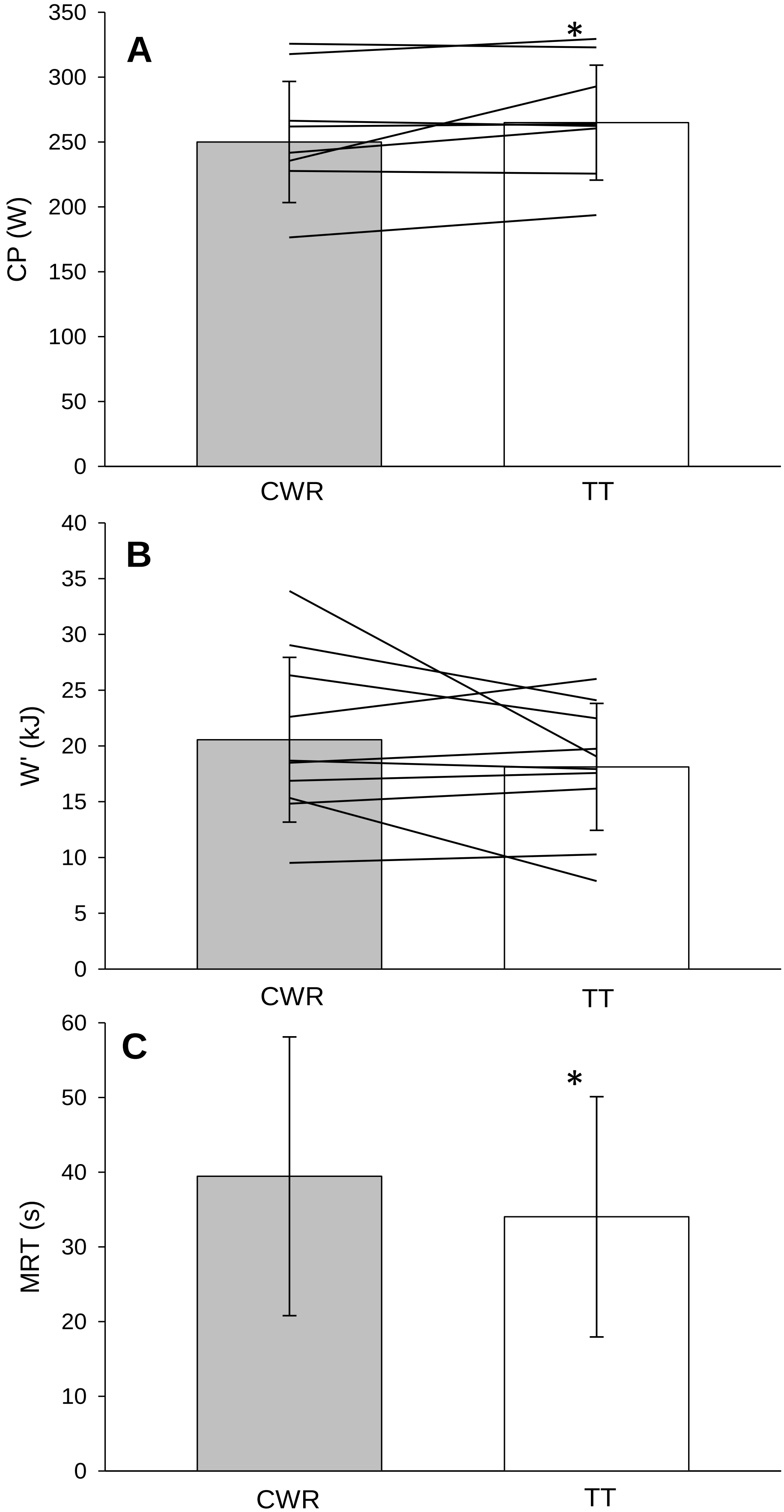
A
CP (W)
350
300
250
200
150
100
50
0
*
CWR
TT
B
W' (kJ)
40
35
30
25
20
15
10
5
0
CWR
TT
C
MRT (s)
60
50
40
30
20
10
0
*
CWR
TT

Figure 2

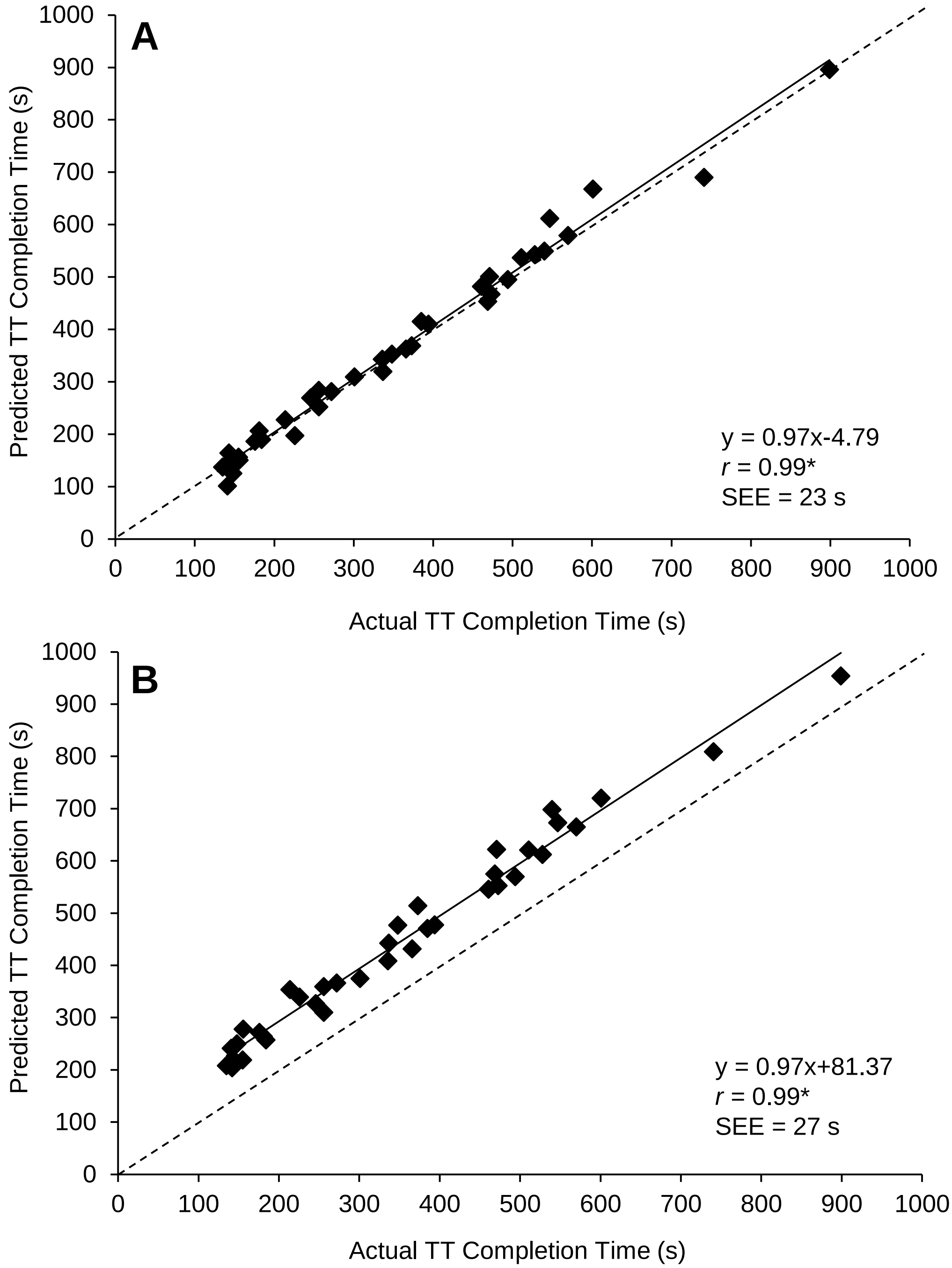
A
Predicted TT Completion Time (s)
Actual TT Completion Time (s)
y = 0.97x-4.79
r = 0.99*
SEE = 23 s
B
Predicted TT Completion Time (s)
Actual TT Completion Time (s)
y = 0.97x+81.37
r = 0.99*
SEE = 27 s

Figure 3